

3	(a)	(i)	Diffraction is the <u>spreading of light waves</u> after it <u>passes through an aperture</u> (of the diffraction grating)	B1
		(ii)	The <u>light from the</u> (adjacent) <u>slits/lines</u> have a <u>constant phase difference</u> between them and same frequency.	B1
		(iii)	Superposition occurs when <u>2 or more light waves of the same kind (or type), meet (or overlap),</u> the <u>resultant displacement is the vector sum of the displacements of the individual light waves</u> at that point.	B1 B1
			*If no reference to light, deduct 1 mark	
	(b)	(i)	Maximum angle for the 2 nd order happens when wavelength is the longest at 650 nm, $d \sin \theta = n\lambda$ $\frac{1 \times 10^{-3}}{500} \sin \theta = 2(650 \times 10^{-9})$ M1 $\theta = 40.5^\circ$ A1	
		(ii)	$d \sin \theta = n\lambda$ $\frac{1 \times 10^{-3}}{500} \sin \theta = 3(350 \times 10^{-9})$ $\theta = 31.7^\circ$ B1	
		(iii)	This means that there is an <u>overlap of fringes</u> between the 2 nd order maxima and 3 rd order minima for white light and is difficult to view fringes of specific wavelengths since they are mixed up.	B1
		(iv)	250 lines per mm means d has increased by 2 times $d \sin \theta = n\lambda$ $n_{\max} \leq \frac{d}{\lambda}$ Slit separation d is increased by 2 times (or doubled), the <u>average separation of the fringes decreases</u> or <u>angle θ for each order is halved</u> . This means <u>maximum order of fringes n is increased by up to 2 times</u> (or doubled).	B1 B1

4	(a)	(i)	N	3×10^{22}	
---	-----	-----	-----	--------------------	--